

L3.2 Adversarial Attacks Quiz

1. What is distribution shift in supervised machine learning?

- A) The inference-data distribution differs from the training-data distribution
- B) The model always receives exactly the training data
- C) The optimizer changes from gradient descent to random search
- D) The class labels are alphabetically reordered

ANS:

2. In the panda adversarial-example demonstration, what does the small perturbation cause the model to predict with high confidence?

- A) A stop sign
- B) A gibbon
- C) A pedestrian
- D) A panda with lower resolution

ANS:

3. What is the main purpose of constraining the norm of an adversarial perturbation?

- A) To guarantee a larger neural network
- B) To maximize the image resolution
- C) To limit the size of the input modification
- D) To remove all model gradients

ANS:

4. Which perturbation pattern is most closely associated with changing every pixel by a small amount?

- A) Changing only the class label
- B) Changing one pixel by an unlimited amount
- C) Deleting the entire image
- D) An L-infinity-bounded perturbation

ANS:

5. What objective is typically maximized when generating an untargeted adversarial example by local search?

- A) The model loss for the correct class
- B) The training-set size
- C) The number of output classes
- D) The image dimensions

ANS:

6. What does Projected Gradient Descent do after taking a gradient step?

- A) It permanently enlarges the permitted perturbation region
- B) It projects the result back into the allowed perturbation set
- C) It retrains the classifier from scratch
- D) It removes every modified pixel

ANS:

7. Which information does FGSM use to determine the perturbation direction?

- A) The sign of the input image
- B) A randomly selected class label
- C) The sign of the gradient of the loss with respect to the input
- D) Only the model's training accuracy

ANS:

8. How does PGD with small steps differ from a single-step FGSM attack?

- A) PGD never uses gradients
- B) PGD changes only model parameters
- C) PGD cannot enforce a perturbation constraint
- D) PGD performs multiple iterative gradient steps with projection

ANS:

9. What is the goal of an untargeted adversarial attack?

- A) To cause any incorrect classification
- B) To force one specific attacker-chosen label only
- C) To improve correct-class confidence
- D) To compress the input image

ANS:

10. What is the goal of a targeted adversarial attack?

- A) To preserve the original prediction
- B) To make the model predict a particular attacker-chosen class
- C) To increase accuracy for every class
- D) To remove the classifier's output layer

ANS:

11. What distinguishes a physically realizable adversarial attack from direct pixel manipulation?

- A) It requires access to every model parameter
- B) It is performed only during model training
- C) It modifies or presents a physical object so that a sensor captures the attack
- D) It cannot affect traffic-sign recognition

ANS:

12. Why must a physical adversarial example be robust to spatial transformations?

- A) To increase the number of classes
- B) To eliminate the need for a camera
- C) To guarantee access to model weights
- D) It may be viewed from different distances, angles, and positions

ANS:

13. What knowledge is assumed in a white-box adversarial attack?

- A) The neural-network model parameters
- B) Only the vehicle's paint color
- C) No information about the model or outputs
- D) Only the number of road lanes

ANS:

14. If an attacker has the target model's training dataset but not its parameters, what black-box strategy is described?

- A) Disable all input sensors
- B) Train a proxy white-box model and generate attacks against it
- C) Replace the target model's hardware
- D) Use only naturally occurring distribution shifts

ANS:

15. How can an attacker collect data for a proxy model when the target training dataset is unavailable?

- A) By reading hidden model parameters directly
- B) By ignoring the target model
- C) By querying the target service to collect input-output pairs
- D) By changing the target's loss function

ANS:

16. What does the YOLO-to-Faster-RCNN example demonstrate?

- A) Adversarial examples never transfer between models
- B) Only digital attacks affect object detectors
- C) A proxy model must be identical to the target model
- D) Physical adversarial examples may transfer from a proxy model to a black-box model

ANS:

17. What is a phantom attack against an ADAS?

- A) A depthless displayed or projected image intended to be treated as a real 3D object
- B) A mechanical failure of the steering actuator
- C) A hidden replacement of the vehicle's processor
- D) A normal reflection that never triggers a response

ANS:

18. Which delivery mechanism for a phantom attack is mentioned in the slides?

- A) A wired keyboard inside the vehicle
- B) A drone-mounted projector or a hacked roadside digital billboard
- C) A replacement tire
- D) A paper map placed in the trunk

ANS:

19. What training approach is used to improve robustness against adversarial inputs?

- A) Training only on empty images
- B) Removing the loss function
- C) Including adversarially generated examples in the training optimization
- D) Freezing every model parameter before training

ANS:

20. According to the slides, why is a robust model harder to attack with small gradient steps?

- A) It has no classification output
- B) It always rejects every input
- C) It uses an infinitely large training set
- D) It has a smoother loss surface

ANS: